Early Identification of High-Aptitude Students from Online Engagement Behavior

Christian Tonny
Introduction to Machine Learning, Summative Project
African Leadership University
GitHub repository. https://github.com/irachrist1/summative-intro-to-ml
Demo video. https://youtu.be/C0IWh0AoSD8

1. Introduction

Talented students in under-resourced schools are easy to lose. A teacher facing a class of sixty, with no specialist training in gifted education and no diagnostic tools beyond end-of-term exams, has almost no way to notice that one quiet student in the third row is capable of far more than the curriculum is asking of her. Research from South African primary schools found that gifted learners were among the most neglected groups in the classroom, and that no deliberate drive to identify or support them existed in the education authorities' agenda [1]. The machine learning literature on African education has a similar blind spot. Most predictive work targets dropout and failure [2]. Almost none of it asks the opposite question, which students are likely to excel, and can we find them early enough to do something about it?

That is the question this project takes on. My problem statement is the early identification of subject-specific academic strengths in students from underserved communities, so that teachers without specialist training can direct support toward high-aptitude students before the opportunity passes. The word early carries most of the weight. A prediction made after final exams is an autopsy. A prediction made in the first month of a course is something a teacher can act on.

I built the project on the Open University Learning Analytics Dataset (OULAD), a public dataset covering 32,593 student-course records from a UK distance university, with demographics, assessment results, and over ten million rows of clickstream activity from the virtual learning environment [3]. Three properties made it the right choice. It records each student's neighbourhood deprivation decile, so socioeconomic disadvantage is measured rather than assumed. It captures observable engagement behaviour at scale, which is the kind of evidence my argument says teachers can use. And it is large enough to train neural networks honestly, which the alternative datasets I profiled (480 to 6,607 rows, two of them synthetic) were not. One disclosure for transparency. I used OULAD in a previous course for a regression task; this project asks a different question with a different task type and a feature set I engineered from scratch.

The technical core is a comparison between traditional machine learning in Scikit-learn and deep learning in TensorFlow, carried through eight systematically varied experiments. Each model sees only the first 30 days of behaviour, roughly the first four weeks of a course, and predicts the final outcome among Distinction, Pass, Fail and Withdrawn. The experiments move from a linear baseline through tuned ensembles and regularised neural networks to a binary reformulation that asks the mission question directly. Can we spot the future Distinction students at day 30?

2. Literature Review

Educational data mining and learning analytics grew from a niche topic into a mature field over the past two decades. Romero and Ventura's updated survey traces this growth and catalogues the field's standard toolkit, classification and regression for performance prediction, clustering for learner profiling, and sequence mining for behaviour analysis [4]. Their survey also notes a recurring weakness that motivates my project. Most published models are evaluated retrospectively on full-course data, which limits their practical use for intervention. A model that needs the whole semester cannot help anyone during it.

The performance-prediction line of work that this project extends starts with Cortez and Silva, who predicted secondary school grades in two Portuguese schools from demographic and social attributes [5]. Their most useful finding was a negative one. Prediction was accurate when prior grades were included and dropped sharply without them, which shows how much of the published accuracy in this area comes from grade-on-grade correlation rather than from anything a teacher could observe. Amrieh, Hamtini and Aljarah took a different route with the xAPI dataset, using behavioural features such as raised hands, resource visits and discussion participation, and reported that ensembles over these behavioural features improved accuracy by around 20 percentage points compared to demographics alone [6]. That result is the seed of my feature strategy. Engagement behaviour carries signal that background variables do not.

Early warning systems put these models in front of educators. The best known is Course Signals at Purdue, which combined grades, effort and prior history into a traffic-light indicator shown to students and instructors, and reported meaningful retention gains in courses that used it [7]. Hlosta, Zdrahal and Zendulka brought the early-prediction idea to OULAD itself with their Ouroboros system, which trains on data from the currently running course so that predictions are available within weeks of the start [8]. Both systems share a framing worth questioning. They exist to find students at risk of failing. The machinery of early prediction is never pointed at the other tail of the distribution, the students who could excel with attention. I read this as a gap, and it is the one this project fills.

Two OULAD studies anchor my modelling choices. Tomasevic, Gvozdenovic and Vranes compared eight supervised techniques on OULAD exam prediction and found artificial neural networks strongest when engagement and past assessment data were combined [9]. Waheed and colleagues built deep feed-forward models on OULAD clickstream slices and reported that deep learning edged out logistic regression and SVM baselines [10]. Both studies, however, drew on observation windows that extend well into the course, and both collapsed the problem to pass versus fail or at-risk versus safe. Neither tried to recover the four-class outcome from the first month alone, and neither treated Distinction as the class of interest. Their accuracy figures are therefore not comparable to mine, and the difference is the point. I am deliberately working in the harder, earlier, less-studied corner of the problem.

On the question of whether deep learning should be expected to win here at all, the tabular-data literature urges caution. Shwartz-Ziv and Armon examined the deep architectures claimed to beat gradient boosting on tabular tasks and found the claims did not survive evaluation on neutral datasets [11]. Grinsztajn, Oyallon and Varoquaux reached the same conclusion across a 45-dataset benchmark and offered reasons, tree ensembles handle irregular feature interactions and uninformative features better than networks on medium-sized tabular data [12]. My results, as Section 5 discusses, agree with them almost embarrassingly well.

Finally, the class-imbalance methods I test have their own history. SMOTE, the synthetic minority over-sampling technique of Chawla and colleagues, generates new minority examples by interpolating between

real neighbours and remains the default resampling method two decades on [13]. Dropout, introduced by Srivastava and colleagues, randomly silences units during training to prevent co-adaptation and is the standard first response to an overfitting network [14]. Both are tested directly in my experiments, and one of them loses.

3. Methodology

OULAD is relational. Demographics and final outcomes live in studentInfo, registration dates in studentRegistration, assessment marks in studentAssessment with deadlines and weights in assessments, and every click on every course material in studentVle, described by vle [3]. The clickstream is the heavy table at 10.6 million rows, which I streamed in two-million-row chunks and filtered to clicks dated before day 30 of each course presentation. Clicks with negative dates are real, they happen between registration and the official start, and I kept them because engaging before day zero is exactly the initiative my problem statement cares about.

The defining methodological decision is the observation window. Every feature is computed from the first 30 days only. From the filtered clickstream I aggregated, per student and course, the total clicks, the number of distinct active days, the number of distinct materials touched, and the day of first access. I then added features that distinguish students with identical totals. Weekly click sums for each of the four weeks feed a least-squares slope that captures whether engagement is rising or fading. Clicks are also split across the seven most common activity types, forums, content pages, quizzes and so on, because a student who spends her clicks in the quiz engine is doing something different from one refreshing the homepage. From the assessment tables I took the mean score, the number of submissions, and the mean submission earliness in days before each deadline, for any work submitted inside the window. Registration timing became a days-registered-early feature. I considered normalising click features within each module, so that engagement is judged against course norms rather than absolute volume, but left it as future work to keep the pipeline simple enough to audit.

Students who withdrew before day 30 were excluded, 5,067 records, because predicting an outcome for someone already gone is not early identification of anything. The final table holds 27,526 student-course records with 49 features after encoding, and the four classes settle at roughly 11% Distinction, 45% Pass, 26% Fail and 18% Withdrawn, as Fig. 1 shows. Distinction remains comfortably the smallest class, which shapes the metric choices below.

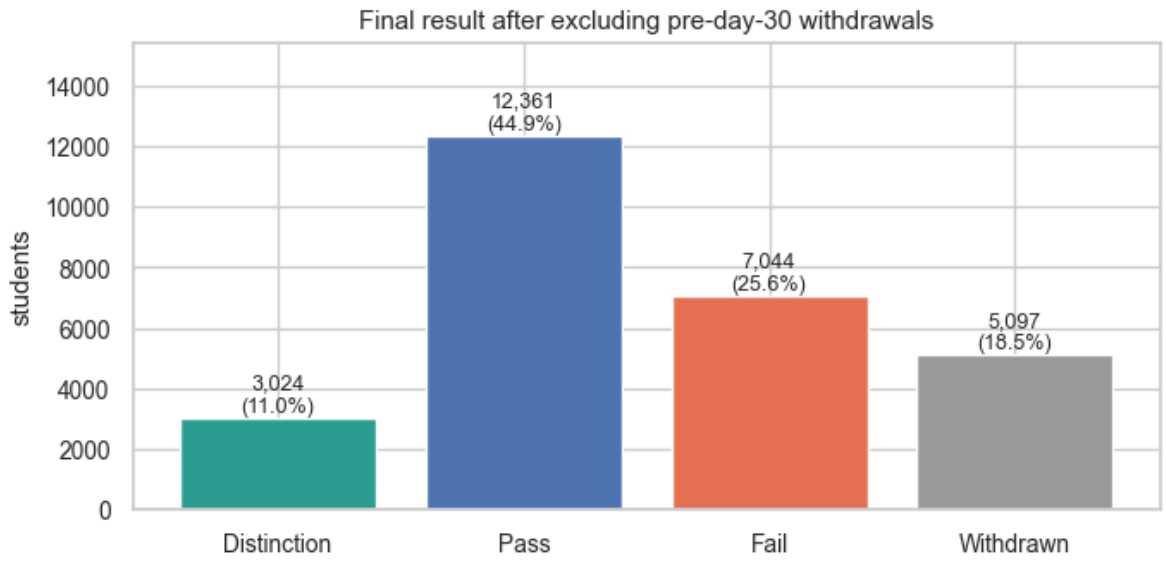


Fig. 1. Distribution of final results after excluding students who withdrew before day 30.

Preprocessing followed the principle that the treatment of a missing value should match the reason it is missing. The UCI export of OULAD marks missing values with a question-mark string, which silently corrupts numeric columns unless caught at read time. Missing click features mean the student never opened the VLE, so they became zeros plus an explicit no-activity flag rather than imputed averages. Missing early-assessment scores usually mean no deadline fell inside the window, so the submission count got zero and the score got the median, with the count signalling when the score is imputed. The deprivation band has 1,111 genuinely unknown values and one mislabelled category, 10-20 missing its percent sign, which I repaired. I imputed the median band and kept a missingness flag, since deleting precisely the students with unknown socioeconomic data seemed a poor choice for this particular project. Ordinal variables, deprivation decile, education level and age band, were encoded as ordered integers; gender, region, disability and module became one-hot columns. Click counts are heavily right-skewed, so all of them passed through a log transform, which costs tree models nothing and helps the linear and neural models considerably. Inputs to those models were standardised with statistics fitted on training data only.

I split the data 70/15/15 into training, validation and test sets, stratified on the outcome, with every random seed fixed at 42. The test set is touched exactly once per experiment. The classical models tune through three-fold cross-validation on the combined training and validation data, while the networks use the explicit validation split for early stopping, so both families see the same total data organised the way each method needs.

The eight experiments form a deliberate sequence in which each one answers a question the previous one raised. Experiment 1 is a class-weighted logistic regression, the linear floor. Experiment 2 swaps in a 300-tree random forest to test whether feature interactions matter. Experiment 3 tunes that forest with GridSearchCV over depth, leaf size and feature sampling, scored on macro-F1. Experiment 4 attacks the imbalance directly with SMOTE feeding a histogram gradient-boosting classifier, with resampling confined inside the pipeline so synthetic students never reach evaluation data. Experiments 5 to 7 move to TensorFlow, all trained through tf.data input pipelines with early stopping, model checkpointing and learning-rate reduction on plateau. Experiment 5 is a plain Sequential network of two hidden layers, 128 and 64 units, trained with class weights and deliberately no regularisation, so that overfitting can be observed before it is treated. Experiment 6 adds dropout of 0.3 and an L2 penalty of 1e-4 [13], [14]. Experiment 7 uses the Functional API to build a two-branch network, behavioural features through a 64-

32 branch and demographic features through a 16-unit branch, concatenated and fused. The justification is empirical. The correlation analysis in Fig. 2 shows the two feature groups are nearly orthogonal, so forcing them through shared early layers spends capacity learning a separation I can encode for free. Experiment 8 reframes the task as binary, Distinction against everyone else, using the tuned forest configuration from Experiment 3, because that is the purest version of the talent-spotting question.

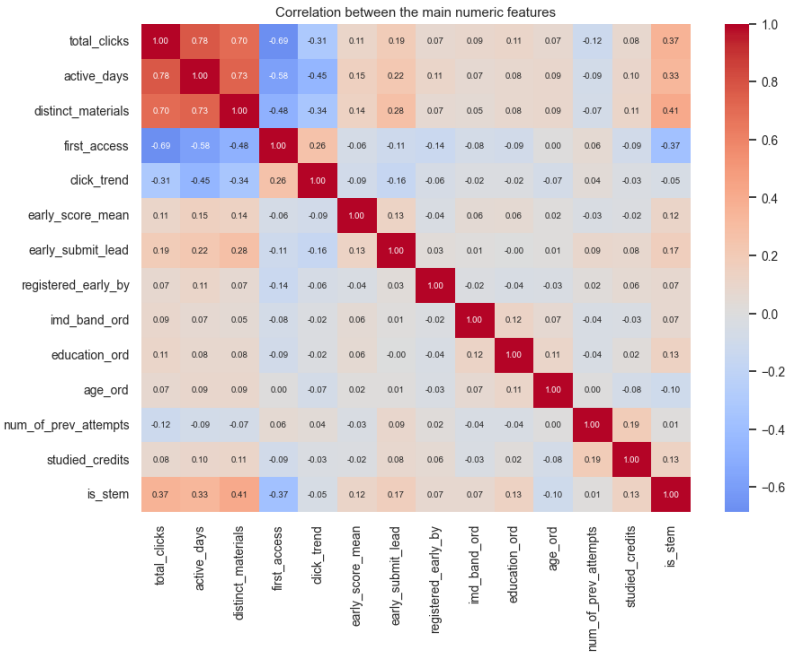


Fig. 2. Correlation between the main numeric features. The behavioural block (top left) is internally correlated but nearly orthogonal to the demographic block.

Accuracy alone would be misleading throughout, which is easiest to see in the binary case, where refusing to ever predict Distinction scores 89%. I therefore report accuracy alongside macro-F1, which averages F1 over classes so each class counts equally, one-versus-rest ROC curves with macro AUC to measure ranking quality independent of any threshold, full per-class precision and recall with confusion matrices, and learning curves for every model. One metric gets special status. Distinction recall is the fraction of genuinely excellent students the model finds, and for this application it is the score that matters most.

4. Results

Table 1 consolidates the eight experiments. All metrics are computed on the held-out test set of 4,129 records; Experiment 8 is binary, so its AUC and recall are not directly comparable to the multi-class rows.

Table 1. Experiment results on the held-out test set.

#	Model	Key settings	Acc.	Macro-F1	Macro AUC	Dist. recall
1	Logistic regression	balanced class weights	0.411	0.409	0.731	0.672
2	Random forest	300 trees, unconstrained	0.526	0.370	0.741	0.099
3	Random forest, tuned	depth 12, leaf 4, 50% features	0.488	0.449	0.744	0.491
4	Gradient boosting + SMOTE	in-pipeline resampling	0.510	0.420	0.726	0.247
5	Sequential NN	128-64, class weights	0.460	0.441	0.736	0.595
6	Sequential NN + dropout/L2	dropout 0.3, L2 1e-4	0.434	0.426	0.742	0.687
7	Functional NN, two-branch	64-32 / 16, concat	0.430	0.423	0.742	0.694

8	Binary, Distinction vs rest	tuned RF settings	0.844	0.664	0.827	0.511

The first thing the table says is that the early-window signal is real, which the raw data had already hinted at. Fig. 3 plots first-month engagement against final outcome, and the medians step downward in order from Distinction to Withdrawn, though with enough overlap between the boxes to warn that no model would separate the classes cleanly. The models agree. Every multi-class model ranks students well above chance, with macro AUC pinned in a narrow band between 0.73 and 0.74. The second thing it says is more interesting. Since ranking quality barely moves across seven very different models, the experiments mostly differ in how they convert the same underlying ranking into decisions, and the column that exposes those differences is Distinction recall.

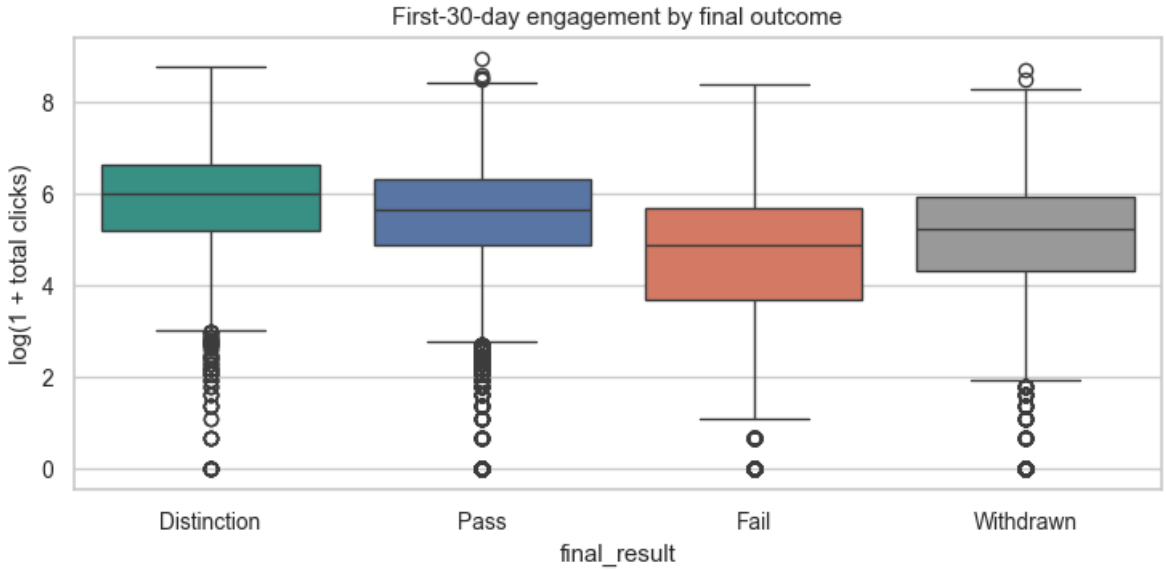


Fig. 3. First-30-day engagement (log of total clicks) by final outcome.

That column tells the central story. The linear baseline finds 67% of Distinction students, but only because balanced class weights make it guess Distinction freely, at 27% precision. Experiment 2 then shows how badly things go when an unconstrained forest meets imbalanced data. Accuracy rises to its peak across all eight experiments, 0.526, while Distinction recall collapses to 0.099. The forest memorises the training set, its leaves become nearly pure, and the class weights lose their influence over prediction. Fig. 4 shows the learning curve, training macro-F1 at 1.0 against a cross-validated score of 0.37, a textbook variance pathology. Experiment 3 treats the diagnosis. The grid search selects a constrained configuration, depth 12, minimum leaf size 4, half the features per split, and Distinction recall recovers to 0.491 while macro-F1 reaches its multi-class best of 0.449. Fig. 5 shows where this model still fails, and the pattern matters. Its errors concentrate in two specific confusions, Distinction with Pass and Withdrawn with Fail.

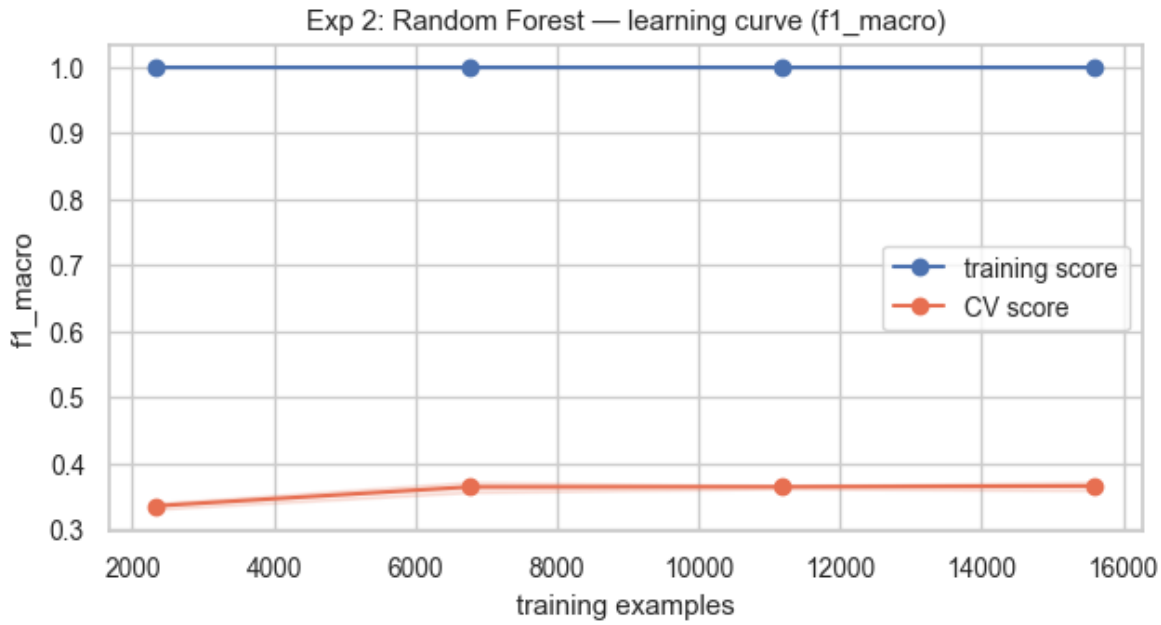


Fig. 4. Learning curve of the unconstrained random forest (Experiment 2), showing the variance pathology.

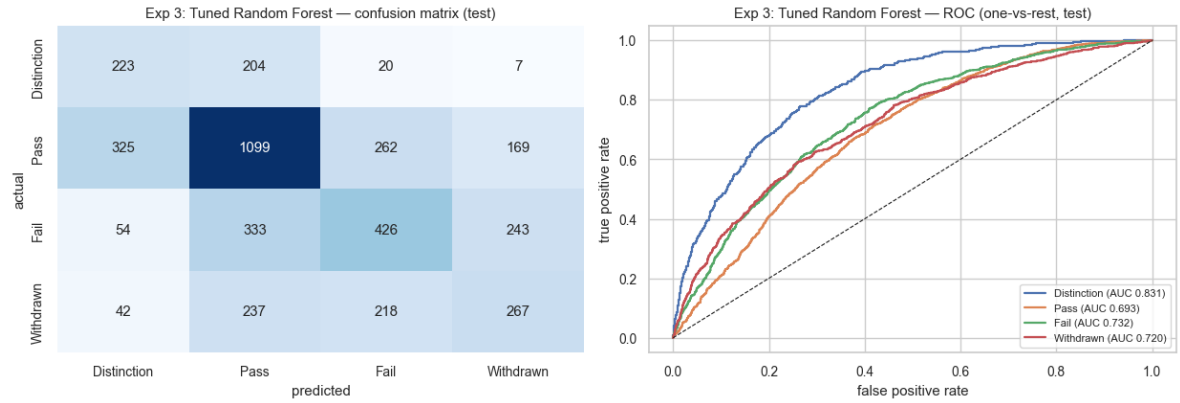


Fig. 5. Confusion matrix and one-vs-rest ROC curves for the tuned random forest (Experiment 3).

Experiment 4 produced the result that surprised me most. I expected SMOTE to deliver the best classical Distinction recall, trading precision away for it. The opposite happened. Recall came out at 0.247, half of what plain class weighting achieved in Experiment 3, while precision rose. Section 5 takes up the explanation. The honest summary here is that resampling lost to reweighting on this dataset, and I am reporting it as the negative result it is.

The neural networks land where the tabular-data literature predicted. As Fig. 6 shows, the unregularised Sequential network overfits visibly, validation loss bottoming out at epoch 9 and drifting upward while training loss keeps sinking; early stopping cut training at epoch 19. Adding dropout and L2 in Experiment 6 changes the picture in Fig. 7, validation loss now tracks training loss and the best epoch moves to 20 of 30. The regularised network gives up about three points of accuracy compared to Experiment 5 and gains in return the best AUC of the deep models, 0.742, and a Distinction recall of 0.687. The two-branch Functional network of Experiment 7 matches that performance with roughly a third of the parameters, 5.7 thousand against 14.9 thousand, but does not exceed it. None of the networks beats the tuned forest's macro-F1.

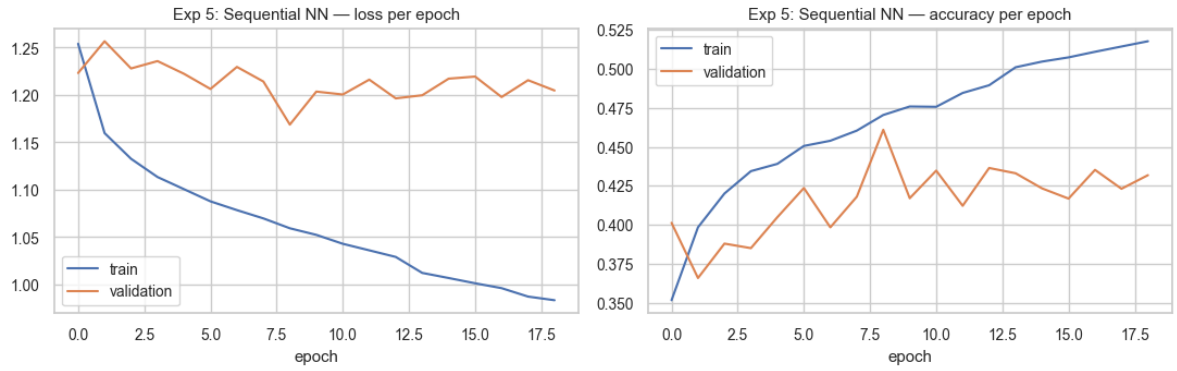


Fig. 6. Training history of the unregularised Sequential network (Experiment 5).

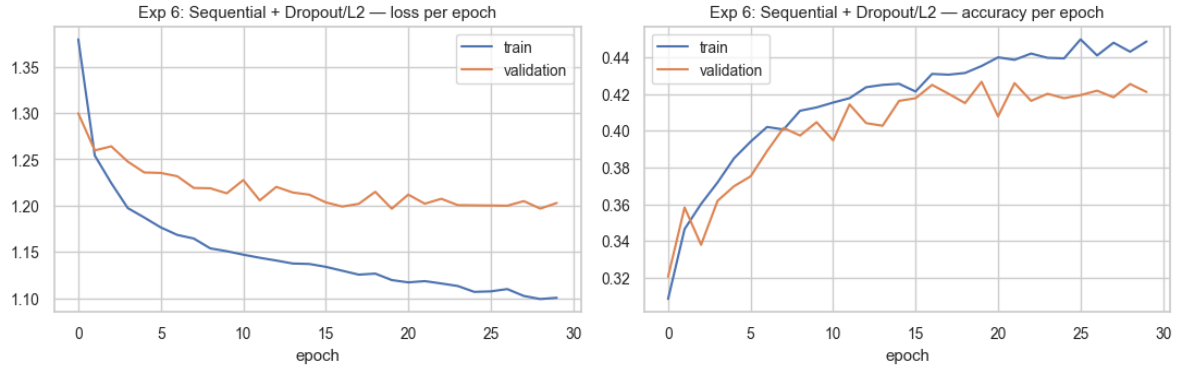


Fig. 7. Training history with dropout and L2 regularisation (Experiment 6).

Experiment 8 ends the sequence by asking the mission question directly, and the reframing pays. Binary AUC reaches 0.827, the strongest ranking result in the project, with the confusion matrix and ROC curve in Fig. 8. At the default threshold the model finds 51% of Distinction students at 35% precision. More usefully, the ROC curve makes the deployment trade explicit, since a school can slide the threshold until the flagged group matches its review capacity and read off the expected hit rate.

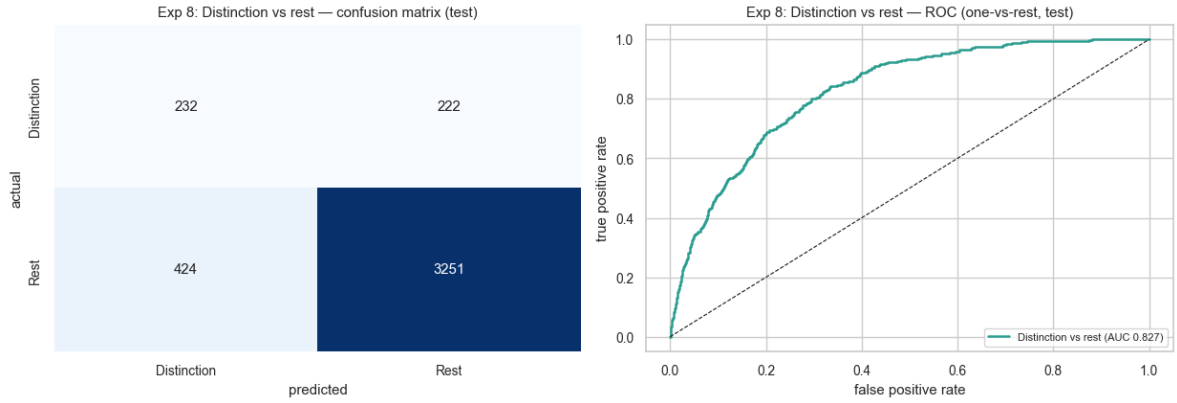


Fig. 8. Confusion matrix and ROC curve for the binary Distinction-versus-rest model (Experiment 8).

5. Discussion

Why did the models behave this way? The clearest lesson sits in the pair of Experiments 2 and 3, where the only change was hyperparameters. An unconstrained forest builds deep trees with nearly pure leaves, and once leaves are pure, the balanced class weights applied during training have almost nothing left to influence at prediction time, so the rare class is simply out-voted. Constraining depth to 12 and leaf size to 4 keeps the leaves mixed, which lets the weights matter again, and recall for the rare class quintuples.

Tuning here was not a percent-chasing exercise. It changed what kind of mistakes the model makes, which for this application is the whole game.

The SMOTE failure deserves its own paragraph because I got my prediction wrong. My best explanation after examining the result is geometric. Distinction and Pass students overlap heavily in this 49-dimensional feature space, especially at day 30 when many eventual Distinctions have not yet pulled away. SMOTE interpolates new minority points between existing neighbours [13], so most synthetic Distinction students are born inside the overlap zone, and the boosting model responds by drawing a careful, precise boundary around the region it can defend rather than casting a wider net. Reweighting, by contrast, reshapes the loss without inventing data and pushed the decision boundary outward. I had treated resampling and reweighting as interchangeable imbalance remedies. They are not, and which one wins appears to depend on how much the minority class overlaps its neighbours.

The deep learning results read as a confirmation of Shwartz-Ziv and Armon and of Grinsztajn and colleagues [11], [12]. I will admit I expected the networks to win before I read those papers closely. With 49 heavily engineered features and 19 thousand training rows, there is little structure left for representation learning to discover that the trees had not already found, and the networks matched but never beat the tuned forest. The two-branch architecture is the subtler case. It was motivated by Fig. 2, which shows the behavioural block and the demographic block correlating internally but barely at all with each other, almost everything in their cross-section below 0.15 in absolute value. Encoding that separation architecturally cost nothing and saved two-thirds of the parameters, but it did not improve performance, and I think the reason is that the demographic branch has little signal to contribute however well it is modelled. That weakness is itself a finding, and a welcome one. What students do in their first month predicts their trajectory far better than where they come from.

The bias-variance story runs through every learning curve in the project. Experiment 1 sits at the high-bias end, training and cross-validation scores converging within 0.01 of each other at a low level, a model that more data cannot help. Experiment 2 sits at the high-variance extreme with its 0.63 gap. Tuning cut the gap roughly in half while raising the validation score, the rare case where variance reduction is free, and the SMOTE pipeline landed between the two forests at about 0.50. On the deep side the same axis appears in epochs rather than tree depth, Fig. 6 against Fig. 7, divergence against convergence, with regularisation buying a longer useful training run. Across both families, every meaningful improvement in this project came from managing variance, never from adding capacity.

The confusion matrices tell one consistent story about errors, and its consistency is the evidence. Across five model families, the same two confusions dominate, Distinction with Pass and Withdrawn with Fail. The first pair differs by degree rather than kind, both are engaged and capable students at day 30, and the boundary between will-pass and will-excel often does not exist yet in the data. The second pair shares a behavioural signature, low and fading engagement, while differing in an administrative outcome. When every architecture from a linear model to a two-branch network fails on the same boundaries, the boundaries belong to the data, not to the models.

The Withdrawn class needs a more honest treatment than I gave it at the start, and this was the limitation I did not anticipate clearly. Withdrawal is not an academic outcome. Students leave courses because of illness, employment, family obligation, finances, none of which a clickstream records. A student withdrawing at day 100 may have been performing at Distinction level until life intervened. Folding Withdrawn into a four-class label treats an administrative event as if it were an ability level, which adds irreducible noise to every multi-class experiment and is part of why Experiment 8, which sidesteps the

issue entirely, produced the cleanest result. If I rebuilt this project I would consider modelling withdrawal separately or removing it from the academic question altogether.

The remaining limitations are about transfer. The data comes from one UK distance university in 2013 and 2014, and engagement patterns of adult online learners will not match those of secondary students in Kigali or Lagos. The module codes are anonymised, so subject-specific strength can only be approximated as STEM against non-STEM, a coarse version of my original aim. The deprivation measure is England-only, leaving 1,111 students imputed and flagged. And the test set holds only 454 Distinction students, so recall estimates for the class I care most about carry real sampling noise. What transfers is the method, not the weights. Early observable behaviour, aggregated simply, ranks talent usefully.

Deployment in an African school context would look different from the research pipeline. Most schools in the contexts I care about run no virtual learning environment, so the features would need analogues that already exist on paper, attendance registers, homework submission timing, participation marks, library or lab usage. The model itself is the cheapest part, a tuned forest trains in seconds on a laptop. The system around it matters more, a calibrated probability rather than a hard label, a flag list sized to the school's actual capacity to respond, and framing that treats a flag as an invitation to look closer, never as a verdict. The asymmetry of errors works in our favour here. A false positive costs a teacher twenty minutes of attention given to a middling student, which is rarely wasted. A false negative is a talented child nobody noticed. I would set the threshold accordingly.

6. Conclusion

This project set out to test whether the first 30 days of observable engagement can identify students likely to excel, and the answer is a qualified yes. Eight experiments spanning logistic regression, tuned and boosted ensembles, and three TensorFlow architectures agree on the essentials. Early behaviour ranks students meaningfully, with every multi-class model reaching macro AUC near 0.74 and the binary Distinction detector of Experiment 8 reaching 0.827. Behavioural features carry the signal and demographics add little, which is the most encouraging possible result for a project about spotting talent in places where background usually decides who gets noticed. The best deployable model is not the most sophisticated one. A tuned random forest answering a binary question beat every neural network I built, in line with what the tabular-data literature led me to expect once I stopped expecting otherwise.

The honest caveats are real. Multi-class accuracy around 0.49 means individual four-way predictions are unreliable, and the value lies in the ranking, not the labels. The Withdrawn class mixes life circumstances into an academic outcome. The data is a decade old, from one institution, in one country, and the Distinction class is small enough that my headline recall figures carry sampling noise. None of this undermines the central finding, but it does scope it.

Future work follows directly. The observation window should be swept from two weeks to ninety days to price earliness properly, since the 30-day choice was made for the problem statement rather than optimised. Click features should be normalised within each module so engagement is judged against course norms. The binary model's probabilities should be calibrated and its threshold chosen against a real review capacity. And the idea needs a pilot with data from an actual African secondary context, using paper-record analogues of the engagement features, to find out how much of the signal survives the transfer.

The mission case stays simple. Every feature that mattered in this project is something a teacher can observe without specialist training, who shows up, who starts early, who touches more of the material, whose effort

is rising. The model is evidence that those observations, even four weeks in, already contain a usable ranking of who might fly. Teachers in under-resourced schools have always lacked tools, not judgement. This is what a cheap tool could look like.

References

[1] M. Oswald and J.-M. de Villiers, "Including the gifted learner: Perceptions of South African teachers and principals," *South African Journal of Education*, vol. 33, no. 1, pp. 1–21, 2013.

[2] N. Mduma, K. Kalegele, and D. Machuve, "A survey of machine learning approaches and techniques for student dropout prediction," *Data Science Journal*, vol. 18, no. 1, art. 14, 2019.

[3] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," *Scientific Data*, vol. 4, art. 170171, 2017.

[4] C. Romero and S. Ventura, "Educational data mining and learning analytics: An updated survey," *WIREs Data Mining and Knowledge Discovery*, vol. 10, no. 3, art. e1355, 2020.

[5] P. Cortez and A. Silva, "Using data mining to predict secondary school student performance," in *Proc. 5th Future Business Technology Conf. (FUBUTEC)*, Porto, Portugal, 2008, pp. 5–12.

[6] E. A. Amrieh, T. Hamtini, and I. Aljarah, "Mining educational data to predict student's academic performance using ensemble methods," *International Journal of Database Theory and Application*, vol. 9, no. 8, pp. 119–136, 2016.

[7] K. E. Arnold and M. D. Pistilli, "Course Signals at Purdue: Using learning analytics to increase student success," in *Proc. 2nd Int. Conf. Learning Analytics and Knowledge (LAK)*, Vancouver, Canada, 2012, pp. 267–270.

[8] M. Hlosta, Z. Zdrahal, and J. Zendulka, "Ouroboros: Early identification of at-risk students without models based on legacy data," in *Proc. 7th Int. Conf. Learning Analytics and Knowledge (LAK)*, Vancouver, Canada, 2017, pp. 6–15.

[9] N. Tomasevic, N. Gvozdenovic, and S. Vranes, "An overview and comparison of supervised data mining techniques for student exam performance prediction," *Computers & Education*, vol. 143, art. 103676, 2020.

[10] H. Waheed, S.-U. Hassan, N. R. Aljohani, J. Hardman, S. Alelyani, and R. Nawaz, "Predicting academic performance of students from VLE big data using deep learning models," *Computers in Human Behavior*, vol. 104, art. 106189, 2020.

[11] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," *Information Fusion*, vol. 81, pp. 84–90, 2022.

[12] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Advances in Neural Information Processing Systems 35 (NeurIPS)*, New Orleans, USA, 2022.

[13] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[14] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *Journal of Machine Learning Research*, vol. 15, pp. 1929–1958, 2014.